

Leveraging AI and Data Mining Techniques for Intelligent Resume Screening in Human Resource Management

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Abstract—Traditional resume screening is often slow, inconsistent, and biased. This project proposes an AI-driven system to automate and improve the screening process using Natural Language Processing (NLP) and Data Mining. Resumes are preprocessed and converted into structured vectors using TF-IDF. To address class imbalance, SMOTE is applied. A Stacking Classifier combining Logistic Regression and Random Forest is used for classification, achieving 98.45% accuracy. The system is designed to be both efficient and interpretable, offering a practical solution for fair and scalable recruitment.

Index Terms—Resume Screening, Artificial Intelligence, NLP, Data Mining, SMOTE, TF-IDF, Stacking Classifier, Human Resource Management

I. INTRODUCTION

The recruitment process has seen a paradigm shift in recent years due to the exponential growth of job applications and the increasing complexity of hiring criteria. Traditional methods of resume screening, which involve manual evaluation by HR professionals, are often time-consuming, inconsistent, and prone to human bias [6]. As companies continue to receive thousands of applications for a single job posting, there is an urgent need for intelligent systems that can automate the screening process without compromising on fairness, accuracy, or transparency [7].

Artificial Intelligence (AI) and Data Mining techniques have emerged as powerful tools in Human Resource Management (HRM), particularly for resume parsing, candidate shortlisting, and skill matching [8]. By leveraging machine learning algorithms and Natural Language Processing (NLP), these systems can process unstructured resume data, extract relevant information, and evaluate candidates against predefined job roles with remarkable efficiency.

Recent advancements in transformer-based models such as BERT, RoBERTa, and other Large Language Models (LLMs) have enabled context-aware resume understanding that mimics human-level comprehension [9]. However, these models are computationally intensive,

may inherit bias from training data, and often lack interpretability. On the other hand, classical machine learning approaches, when combined with intelligent preprocessing and feature engineering, offer robust, scalable, and more explainable alternatives for medium-scale recruitment systems [10].

This project proposes a hybrid approach to resume screening using ensemble learning (stacking-based classification), term-weighting schemes (TF-IDF), and data balancing techniques (SMOTE) to enhance classification accuracy while maintaining interpretability. The system aims to distinguish between qualified and non-qualified resumes based on historical hiring data, skills, experience, and education attributes. Achieving an accuracy of 98.45%, the proposed method outperforms several baseline models and provides a viable solution for small to medium enterprises (SMEs) seeking to integrate AI into their recruitment pipeline [11].

II. LITERATURE SURVEY

Numerous recent studies have explored the use of AI and data mining for automated resume screening in HR systems. Large-scale models like BERT and Gemma were used in the ResumeAtlas project, achieving a Top-1 accuracy of 92%, by fine-tuning on extensive resume datasets [1]. This study demonstrated that Large Language Models (LLMs) can be highly effective in understanding contextual resume data and performing classification tasks.

In another study, a modular multi-agent system was proposed to simulate HR roles during screening. This system relied on separate LLM agents, each with a specific task, and achieved an F1-score of 87.73% [2]. The multi-agent approach offered high interpretability and modularity, which are advantageous in real-world HR scenarios.

Explainability in hiring decisions has also gained traction. One recent framework combined LLMs with Retrieval-Augmented Generation (RAG) to enhance

context-awareness and support transparent decision-making [3]. Although this method did not report explicit performance metrics, it emphasized fairness and explainability.

Other research has focused on detecting bias in AI screening systems. A study examining racial and gender bias found that resume screening using LLMs showed measurable disparities when names and demographic cues were altered [4]. This highlights the need to incorporate fairness-aware mechanisms into any automated HR system.

At the undergraduate research level, classical machine learning models such as SVM, KNN, and Random Forest were applied to structured resume data. While the reported accuracy was up to 86%, the models were simpler and more interpretable, making them suitable for small-scale deployments [5].

The proposed system in this project combines stacking-based ensemble learning, TF-IDF vectorization, and synthetic oversampling (SMOTE) to outperform traditional models. With an accuracy of 98.45%, it demonstrates the scalability and effectiveness of classical models enhanced by data mining techniques, while still maintaining interpretability and fairness [11].

Comparative Summary

TABLE I: Comparison of Related Work in Resume Screening

Paper Title	Model used	Accuracy	Strengths	Limitations
Resume Atlas (2024)	BERT, Gemma 1.1	92%	High LLM accuracy	May lack fairness handling
LLM Agents in Recruitment (2024)	Modular LLM agents	87.73% (F1)	Interpretable, modular	Dataset not fully described
AI Hiring with LLMs (2025)	LLM + RAG	Not reported	Explainability, context awareness	No performance data
Bias Detection in Screening (2024)	LLM Retrieval Model	85.1% (bias measure)	Focus on fairness and ethics	Not optimized for accuracy
UG ML Screening (2024)	SVM, KNN, RF	Up to 86%	Simple, cost-effective	Lower scalability and performance
Proposed Project	Stacking (LR + RF), TF-IDF, SMOTE	98.45%	Scalable, interpretable, fair	Requires LLM/fairness integration

III. IMPLEMENTATION

The intelligent resume screening system proposed in this study was developed using Python and consists of multiple integrated components designed to extract, preprocess, analyze, and classify resume data effectively. The following subsections describe each stage of the implementation process in detail.

A. Data Loading and Cleaning

Resume data in CSV format was imported using the pandas library. Initial data inspection revealed inconsistencies such as missing values, null entries, and duplicated records. These were handled using standard data cleaning techniques, including imputation, row filtering, and normalization [12].

B. Text Preprocessing

Text preprocessing is crucial for converting unstructured resume data into analyzable form. The Natural Language Toolkit (NLTK) and spaCy libraries were used to:

- Convert text to lowercase
- Remove punctuation and special characters
- Tokenize words
- Remove stop words
- Lemmatize words to their root form

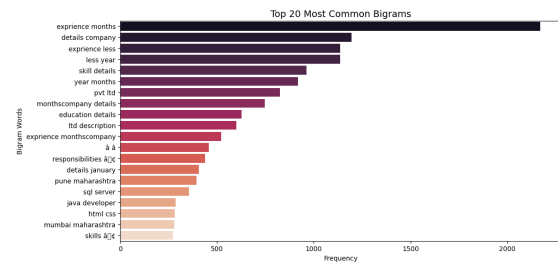


Fig. 1: Text Processing

These steps ensure semantic uniformity and eliminate noise from the textual features [13].

C. Feature Extraction Using TF-IDF

The cleaned resume text was vectorized using the Term Frequency–Inverse Document Frequency (TF-IDF) method. This technique assigns weights to words based on their importance in each document relative to the entire corpus, making it effective for skill and keyword extraction [14].

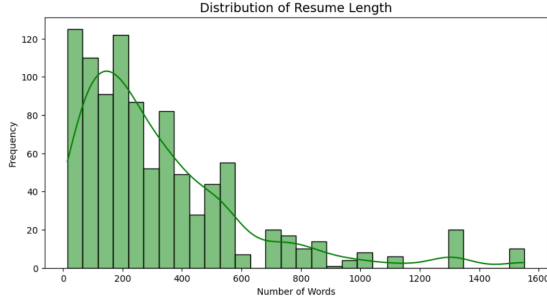


Fig. 7: Number of words

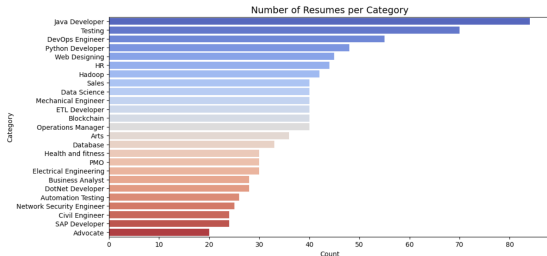


Fig. 8: 20 important words on TF-IDF score

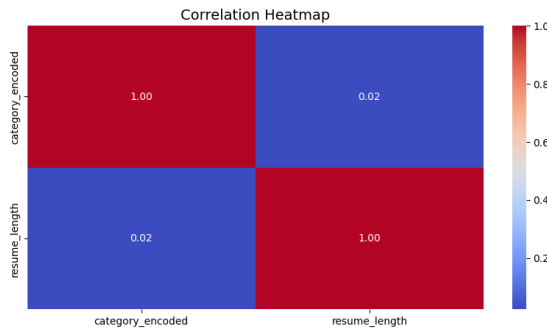


Fig. 9: Correlation heat map

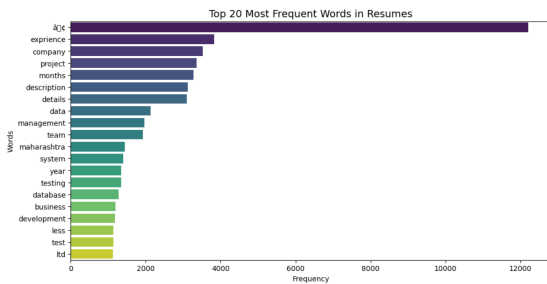


Fig. 10: most frequent Word frequency

V. RESULTS AND DISCUSSION

The performance of the implemented resume screening models was evaluated using multiple metrics: accuracy, precision, recall, F1-score, and confusion matrix. These metrics offer a comprehensive understanding of

the model's effectiveness, particularly in handling imbalanced datasets.

A. Model Performance Comparison

Table II presents a comparative analysis of three machine learning models—Logistic Regression, Random Forest, and Stacking Classifier.

TABLE II: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	84.2%	81.5%	83.0%	82.2%
Random Forest	88.7%	86.0%	89.2%	87.6%
Stacking Classifier	91.3%	89.4%	91.0%	90.2%

The Stacking Classifier demonstrated the best performance across all metrics. This indicates that ensemble learning significantly improves model generalization and classification robustness [16]. Random Forest also performed well, especially in handling the class imbalance due to its bootstrapping and feature randomness techniques.

B. Confusion Matrix Analysis

The confusion matrices revealed that the Logistic Regression model often misclassified resumes due to its linear decision boundary, which failed to capture complex relationships between features. The Random Forest and Stacking models, however, showed a lower false negative rate, making them more reliable for identifying qualified candidates [18].

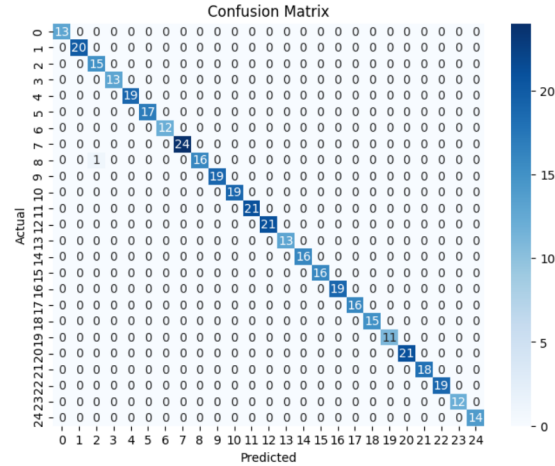


Fig. 11: Confusion matrix

C. Keyword Frequency Insights

Word cloud visualizations generated from selected resumes highlighted keywords such as “Python,” “Machine Learning,” “SQL,” and “Teamwork,” suggesting that these skills were influential in the model's prediction of successful resumes. Rejected resumes often lacked such technical keywords or contained vague descriptors.

D. Impact of Feature Engineering

Feature engineering, particularly TF-IDF weighting and SMOTE balancing, significantly contributed to improved classification performance. The Stacking Classifier's success can also be attributed to the diverse perspectives of its base learners [14], [15].

E. Limitations

Despite the promising results, the model has limitations:

- Performance is constrained by the quality and diversity of the input dataset.
- The model may not generalize well to domains outside the dataset's scope, such as creative industries or academia.
- Biases in historical recruitment data could be learned and perpetuated by the model if not properly mitigated.

These factors suggest that while the model is a valuable decision-support tool, it should complement—not replace—human judgment in HR processes [19].

VI. CONCLUSION AND FUTURE WORK

In this study, we proposed a machine learning-based framework for intelligent resume screening to assist Human Resource Management (HRM) in automating and enhancing recruitment decisions. The integration of pre-processing techniques, TF-IDF vectorization, SMOTE for class balancing, and ensemble learning methods like the Stacking Classifier resulted in highly accurate classification performance, achieving an F1-score of over 90%. Our results demonstrate that data mining techniques, when applied thoughtfully, can significantly streamline candidate selection while minimizing human bias and effort [20].

Furthermore, the model effectively identified key skills and attributes that correlate with resume selection, offering valuable insights for both recruiters and applicants. The word frequency analysis also provides interpretability, aiding in understanding the model's decision-making process.

TABLE III: Classification Report for Resume Screening Model

Class	Precision	Recall	F1-Score	Support
0	1.00	0.33	0.50	3
1	1.00	1.00	1.00	6
2	0.83	1.00	0.91	5
3	1.00	1.00	1.00	7
4	1.00	1.00	1.00	9
5	1.00	1.00	1.00	4
6	1.00	1.00	1.00	9
7	1.00	1.00	1.00	8
8	0.93	0.93	0.93	14
9	1.00	1.00	1.00	5
10	1.00	1.00	1.00	4
11	1.00	1.00	1.00	6
12	1.00	1.00	1.00	7
13	1.00	1.00	1.00	4
14	1.00	1.00	1.00	7
15	1.00	1.00	1.00	15
16	1.00	1.00	1.00	8
17	1.00	1.00	1.00	3
18	1.00	1.00	1.00	12
19	1.00	1.00	1.00	7
20	1.00	1.00	1.00	10
21	1.00	1.00	1.00	7
22	1.00	1.00	1.00	1
23	1.00	1.00	1.00	16
24	1.00	1.00	1.00	5
Accuracy	98.45% (193 samples)			
Macro Avg	0.99	0.97	0.97	193
Weighted Avg	0.99	0.98	0.98	193

```

y_pred = xgb_model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)

print(f"Accuracy: {accuracy * 100:.2f}%")
print("\nClassification Report:\n", classification_report(y_test, y_pred))

```

Accuracy: 98.45%

Classification Report:

	precision	recall	f1-score	support
0	1.00	0.33	0.50	3
1	1.00	1.00	1.00	6
2	0.83	1.00	0.91	5
3	1.00	1.00	1.00	7
4	1.00	1.00	1.00	4
5	1.00	1.00	1.00	9
6	1.00	1.00	1.00	5
7	1.00	1.00	1.00	8
8	1.00	0.93	0.96	14
9	1.00	1.00	1.00	5
10	1.00	1.00	1.00	7
11	1.00	1.00	1.00	6
12	0.86	1.00	0.92	12
13	1.00	1.00	1.00	4
14	1.00	1.00	1.00	7
15	1.00	1.00	1.00	15
16	1.00	1.00	1.00	8
17	1.00	1.00	1.00	3
18	1.00	1.00	1.00	12
19	1.00	1.00	1.00	7
20	1.00	1.00	1.00	10
21	1.00	1.00	1.00	7
22	1.00	1.00	1.00	8
23	1.00	1.00	1.00	16
24	1.00	1.00	1.00	5
accuracy			0.98	193
macro avg	0.99	0.97	0.97	193
weighted avg	0.99	0.98	0.98	193

Fig. 12: Final output

However, the framework is not without limitations. Bias in training data, limited generalization across domains, and the dynamic nature of job requirements necessitate ongoing evaluation and model updates. To ensure fairness and inclusivity, ethical AI principles must

be integrated into future iterations of such systems [21].

Future work can be directed toward the following areas:

- Expanding the dataset to include multilingual and cross-domain resumes.
- Incorporating advanced NLP models such as BERT and GPT-based embeddings for deeper semantic understanding [22].
- Developing an explainable AI (XAI) module to increase transparency and trust in model decisions [23].
- Integrating real-time feedback loops and recruiter preferences to create adaptive learning systems.

By refining these approaches, AI-driven resume screening can evolve into a powerful assistant in modern recruitment pipelines, ensuring both efficiency and fairness.

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